

CHAPTER TEST WITH ANSWERS

Chapter 8 Analytical techniques

Time permitted: 50 minutes

	Section	Number of questions	Marks available
A	Multiple choice	15	15
B	Short answer	5	15
	Total	20	30

Scale:

A+	29–30	A	26–28	B	23–25	C	19–22	D	15–18	E	9–14	UG	0–8
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Section A Multiple choice (15 marks)

Section A consists of 15 questions, each worth one mark. Each question has only one correct answer. Circle the correct answer. Attempt all questions. Marks will not be deducted for incorrect answers. You are advised to spend no more than 15 minutes on this section.

- X-ray crystallography is an analytical technique that:
 - fires X-rays at crystals to determine their age.
 - detects diffraction patterns from atoms within a crystal.*
 - uses X-rays to find crystalline structures.
 - diffracts X-rays from damaging precious crystals.
- Analytical techniques help us determine chemical structures. Which of the following is true?
 - X-ray crystallography determines bond angles within crystals.*
 - Nuclear magnetic resonance determines bond angles within crystals.
 - Mass spectrometry uses magnetic fields to help determine structures of molecule.
 - Infrared spectroscopy uses light rays to determine structures of molecules.
- In the electromagnetic spectrum the energy of a wave is inversely proportional to wavelength:
 - so that smaller wavelengths have less energy.
 - so that smaller wavelengths have more energy.*
 - so that larger wavelengths have more energy.
 - so that longer wavelengths have more energy.

- 4 Mass spectrometry knocks off parts of a molecule that are:
- A deflected and identified as a result of their mass.
 - B ions from the molecule.
 - C detected by their charge.
 - Ⓓ *deflected and identified as a result of mass and charge.*
- 5 Chlorine has a relative atomic mass as a result of:
- Ⓐ *the relative abundance of isotopes.*
 - B the relative mass of the particles in one atom.
 - C the percentage of chlorine in the atmosphere.
 - D the low proportion of chlorine in nature.
- 6 In mass spectrometry the highest peak in the spectrum:
- A reveals the free radicals present.
 - B determines the parent molecule.
 - Ⓒ *is the most stable fragment of the molecule.*
 - D shows the most deflected fragment.
- 7 Nuclear magnetic resonance (NMR) determines structures by:
- A the number of hydrogen molecules present.
 - Ⓑ *the environment of hydrogen atoms in the molecule.*
 - C how many hydrogen molecules are attached to the same carbon atom.
 - D shielding electrons on the carbon atoms.
- 8 TMS in NMR is Tetramethylsilane, which is a standard because it:
- A has the most shielded electrons on the Si atom.
 - Ⓑ *has the least shielded electrons on the H atoms.*
 - C has the least shielded electrons on the Si atoms.
 - D has the most shielded electrons on the H atoms.
- 9 Infrared spectroscopy shows:
- A emission bands of excited molecular bonds stretching.
 - B absorption bands due to energy absorption by bonds rotating.
 - Ⓒ *absorption bands due to energy absorption by bonds stretching.*
 - D emission bands of excited molecular bonds rotating.

10 UV visible spectroscopy uses standard solutions to:

- Ⓐ *produce a calibration curve.*
- B determine where absorption occurs.
- C determine unknown concentrations.
- D calculate and identify the unknown compound.

11 Carbon-13 NMR spectroscopy focuses on:

- A hydrogen nuclei.
- B carbon nuclei.
- C electronegativity of C.
- Ⓓ *electrons held by C atoms.*

12 Infrared spectroscopy is caused by absorbance of infrared radiation by:

- A bonds stretching in a compound.
- B atoms in compound.
- C bonds bending in a compound.
- Ⓓ *bonds bending and stretching in a compound.*

13 Mass spectrometers rely on:

- A ionisation, acceleration and detection only.
- Ⓑ *ionisation, acceleration, deflection and detection.*
- C ionisation, deflection and detection.
- D ionisation, acceleration and deflection.

14 Mass spectrometry requires a vacuum:

- A to remove any unwanted fragments.
- Ⓑ *so particles do not hit air particles.*
- C to remove free radicals.
- D so ions formed are not repelled by any other gases.

15 Deflection occurring within a mass spectrometer depends on:

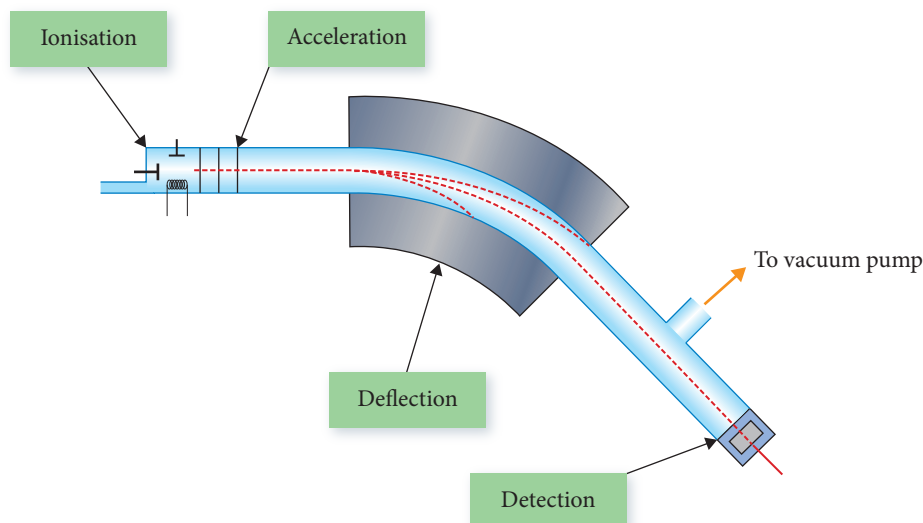
- Ⓐ *mass charge ratio.*
- B mass of particle.
- C charge of particle.
- D ionisation of particle by electrons.

Section B Short answer (15 marks)

Section B consists of five questions. Write your answers in the spaces provided. You are advised to spend 25 minutes on this section.

- 1 a Label the following diagram of a mass spectrometer to show ionisation, detection, deflection and acceleration.

Answer:



(1 mark)

- b Referring to the diagram in part a (on page 4) explain where the sample goes in, how it bends through the machine and where it is detected.

Answer: The sample is ionised by high-energy electrons. The ions are then accelerated to increase their speed, and are then deflected by magnetism, which spreads. (1 mark)

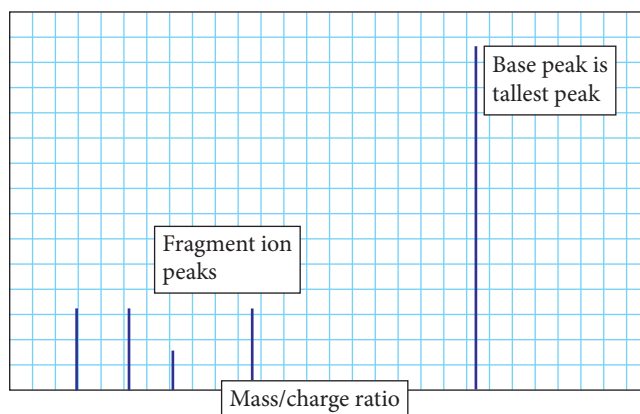
- c Explain mass charge ratio and deflection of particles in the mass spectrometer.

Answer: Mass charge ratio affects deflection of ions due to mass charge ratio (the more it is charged, the more it is deflected and the heavier it is, the less it deflects). For example, a mass of 58 and charge of +2 have a mass charge ratio of 28; a particle of mass 28 and charge of +1 will also have a mass charge ratio of 28. (1 mark)

(= 3 marks total)

- 2 Label the following mass spectrometry fragmentation graph, including axes, base peak, molecular ion peak and fragment ion peaks.

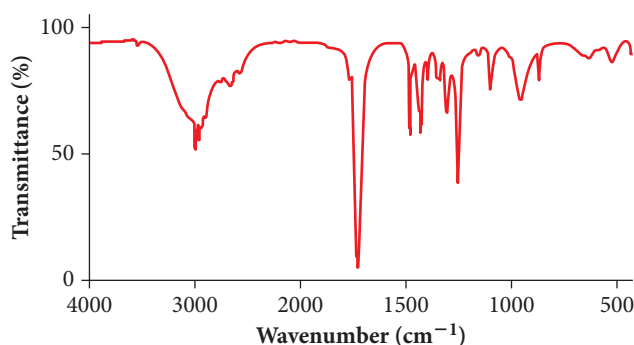
Answer:



(3 marks)

(= 3 marks total)

- 3 Identify the groups that are revealed in the following spectrum, reveal why this information is used in conjunction with mass spectrometry and NMR data in identifying the compound it is.



Answer: 3 C-H stretch 2850 to 3100 and C=O stretch at 1750

NMR determines the molecules C-H backbone and IR spectrometry determines its functional groups. So the two techniques reveal the molecule.

(= 3 marks total)

- 4 Explain how UV spectroscopy works and how it is used by scientists in the laboratory.

Answer: UV can determine if a substance is in the solution of a specific absorption band. UV absorption spectroscopy finds the concentrations of known solutions of a substance. An unknown solution of the compound can then have its concentration determined by comparing the absorption of it compared to the known solutions. It works as the electrons absorb energy and are excited into higher energy levels at signature wavelengths that are available in tables.

(= 3 marks total)

- 5 Explain how X-ray diffraction analysis of crystals occurs and how it is used to determine structure of compounds.

Answer: X-ray crystallography works by shining X-rays onto a crystal of the compound. The X-rays are diffracted by high-electron density areas in the molecule. This produces a diffraction map of the molecule, which is then analysed and interpreted by computer analysis. This reveals bond lengths and angles within the structure.

(= 3 marks total)